Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/24/2018
PI Name:	Boerma, Marjan Ph.D.		
Project Title:	Center for Research on Cardiac, Vascular, and Acute Effects of Space Radiation		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mboerma@uams.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	501-686-6599
Organization Name:	University of Arkansas, Little Rock		
PI Address 1:	4301 W. Markham Street, Slot 522-10		
PI Address 2:	Slot 522-10		
PI Web Page:			
City:	Little Rock	**State:**	AR
Zip Code:	72205-7101	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 NSBRI-RFA-13-02 Center for Space Radiation Research (CSRR)
Start Date:	06/01/2014	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Xiao M.D. (Loma Linda University) Hauer-Jensen, Martin M.D., Ph.D. (University of Arkansas for Medical Sciences) Kodell, Ralph Ph.D. (University of Arkansas for Medical Sciences) Koturbash, Igor M.D., Ph.D. (University of Arkansas for Medical Sciences) Tackett, Alan Ph.D. (University of Arkansas for Medical Sciences) Nelson, Gregory Ph.D. (Loma Linda University)		
Grant/Contract No.:	NCC 9-58-RE03701		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Center for Research on Cardiac, Vascular, and Acute Effects of Space Radiation consisted of teams of investigators from the University of Arkansas for Medical Sciences (UAMS), Loma Linda University (LLU), the University of Arizona (UAZ), and Georgetown University. We used multiple animal models to characterize acute effects of protons at doses lower than addressed in previous animal studies, and experiments involved exposure of animals and cell cultures to protons and heavy ions to examine degenerative cardiovascular effects. We pursued the following Specific Aims: 1) Define acute effects of low-dose proton irradiation on the hematopoietic system, skin, heart, and retina; 2) Examine acute effects of low-dose protons in combination with modeled microgravity; 3) Evaluate acute effects of protons on the adaptive immune response; 4) Determine effects of heavy ion and proton irradiation on cardiac and vascular function and structure; 5) Identify acute and chronic biomarkers of cardiovascular dysfunction after particle irradiation; 6) Elucidate the role of metabolic and epigenetic changes in the cardiovascular response to radiation; 7) Determine effects of radiation dose and quality on endothelial cell functional phenotype; and 8) Test whether gamma-tocotrienol protects against cardiovascular effects of particle irradiation. Key studies and findings under each Specific Aim: 1) Male C57BL/6 mice were irradiated at 6 months of age with a fully modulated 150 MeV proton beam to mimic a solar particle event (SPE) or cobalt-60 gamma-rays at doses of 0.1 – 0.5 Gy, and blood and spleen cells were quantified with a hematology analyzer and flow cytometry. Doses =0.1 Gy of both protons and gamma-rays caused a decrease in blood cell counts at 60 hours and 4 days, followed by a gradual repopulation after 1 week. These studies contribute to the determination of dose threshold for acute effects. 2) Mice were subjected to hindlimb unloading to remove mechanical loads from the posterior musculoskeletal system and generate a cephalic fluid shift, both of which occur in microgravity. Thus, animals were hindlimb unloaded for 7 days, irradiated at 0.5 Gy with modulated 150 MeV protons, and hindlimb unloaded for an additional 7 days. The most severe decrease in whole blood cell and lymphocyte counts was seen in animals exposed to both radiation and unloading (i.e., a significant interaction was seen between radiation and hindlimb unloading). Similarly, apoptosis and cellular senescence in the retina were most severe in the combined exposure group. 3) The mouse ear swelling test was used to assess adaptive immunity. Mice were challenged with an injection of FITC, followed 4 days later by modulated 150 MeV proton irradiation at a dose of 0.5 Gy. When mice were re-challenged after 2 weeks with an injection of fluorescein isothiocyanate (FITC) into one of the ears, relative ear swelling was significantly increased in the irradiated group compared with sham-irradiated animals. These results suggest that solar particle event (SPE)-like proton exposures may enhance the adaptive immune response, with potential negative consequences in hypersensitivity reactions and allergies. 4) Male C57BL/6 mice were used to determine cardiac function, tissue structure, and protein expression up to 9 months after exposure to 16O (600 MeV/n, 0.05 – 1 Gy), protons (150 MeV, 0.5 or 1 Gy), or 137Cs gamma-rays (0.5 – 3 Gy). 16O caused small but significant decreases and protons small increases in echocardiographic parameters. All 3 types of radiation induced protein markers of cardiac inflammatory infiltration at =1 Gy. Additionally, male Long Evans rats and male New Zealand White rabbits were exposed to 16O (600 MeV/n, 0.5 Gy) or protons (250 MeV, 0.5 Gy) and followed for 12 months after irradiation. Both rats and rabbits showed a small but significant increase in plasma levels of the injury marker cardiac troponin I (cTnI). In rabbits, this coincided with small increases in ejection fraction as measured with echocardiography. On the other hand, no significant changes were found in cardiac mast cell numbers or collagen deposition in any of the three species. In addition, no changes were seen in the structure of the retinal vasculature in irradiated mice. Altogether, cardiovascular effects of radiation were mild. 5) Proteomics was performed on samples of mouse left ventricle and plasma collected under Specific Aim 4. Pathways that were most commonly identified by Ingenuity Pathway Analysis included mitochondrial dysfunction, the nuclear receptor FXR/RXR pathway, and others. 6) Metabolomics was performed on samples of mouse left ventricle, plasma, urine, and feces collected under Specific Aim 4. Low doses of 16O (0.1 Gy and 0.25 Gy) induced more changes in metabolite profiles than 1 Gy. These profiles suggest inflammation, and oxidative and nitrosative stress. Alterations in one-carbon metabolism and DNA methylation pointed to potential molecular mechanisms by which oxygen ions modify the heart. Some of the changes in fecal metabolites could be attributed to radiation-induced changes in the microbiome. 7) Cultures of mouse and human retinal and cardiac endothelial cells were exposed to protons, heavy ions, and gamma-rays. Alterations in protein expressions were dependent on radiation type, dose, and cell type. In addition, doses =0.1 Gy caused a decrease in tubule network formation when cells were plated on a Matrigel/collagen matrix 6 hours after irradiation, indicative of a reduced capacity for early-onset angiogenesis. 8) Male C57BL/6 mice were exposed to 16O (600 MeV/n, 0.25 Gy) and administered gamma-tocotrienol (s.c., 50 mg/kg) once a day Monday – Friday for 4 weeks after irradiation. When measured 2 weeks after irradiation, gamma-tocotrienol reduced left ventricular protein levels of CD2 and collagen type III, but did not modify CD68 or mast cell tryptase levels. Analysis of echocardiography recordings and tissue samples at remaining post-radiation time points is being finalized.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There is renewed interest in the chronic cardiovascular effects of terrestrial exposures to low doses of ionizing radiation, such as from accidental exposure or medical and diagnostic treatments. This project has assessed chronic effects of low-dose ionizing radiation on heart and vasculature and has started the identification of potential biological mechanisms. These studies will contribute to the general understanding of the cardiovascular effects of low-dose ionizing radiation. The Armed Forces Radiobiology Research Institute (AFRRI) and other government agencies have a longstanding interest in gamma-tocotrienol as a potential countermeasure against radiation from a nuclear attack or accident and has collaborated with several investigators that have also served on the current project. Results obtained from the current studies indirectly support our efforts to develop gamma-tocotrienol as a countermeasure against terrestrial radiation exposure.

Task Progress:	This project used animal models and endothelial cell cultures to characterize acute effects of protons at doses lower than previously studied and examine degenerative cardiovascular effects of protons and oxygen ions. Specific Aims: 1) Define acute hematopoietic effects of low-dose protons; 2) Examine acute effects of protons in combination with simulated microgravity; 3) Evaluate effects of protons on the adaptive immune response; 4) Determine effects of heavy ions and protons on cardiac and vascular function and structure; 5) Identify biomarkers of cardiovascular dysfunction after particle irradiation; 6) Elucidate the role of metabolic and epigenetic changes in the cardiovascular response to radiation; 7) Determine effects of radiation dose and quality on endothelial cell phenotype; 8) Test whether gamma-tocotrienol protects against cardiovascular effects of radiation. Key findings under each aim: 1) A decrease in blood cell counts up to 4 days after protons and gamma-rays at doses =0.1 Gy in mice. These studies contribute to the determination of dose threshold for acute effects; 2) A significant interaction between simulated microgravity and radiation on blood cell count, and apoptosis and cell senescence in the mouse retina; 3) An enhanced adaptive immune response in mice after protons, which may have potential negative consequences in hypersensitivity reactions; 4) Small changes in echocardiography parameters and protein markers of inflammatory infiltration in the heart of proton or oxygen ion exposed mice, rats, and rabbits. Rats and rabbits showed a small but significant increase in plasma cardiac troponin I. Mice showed no changes in retinal vascular structure months after irradiation. Altogether, cardiovascular changes were mild; 5) Proteomics performed on samples of mouse heart and plasma revealed common pathways including mitochondrial dysfunction, the nuclear receptor FXR/RXR pathway, and others; 6) Metabolomics performed on samples of mouse heart, plasma, urine, and feces revealed more changes after low doses of oxygen ions (0.1 Gy, 0.25 Gy) compared to 1 Gy. Alterations in one-carbon metabolism and DNA methylation pointed to potential molecular mechanisms by which oxygen ions modify the heart. Some of the fecal metabolites were attributed to radiation-induced changes in the microbiome; 7) Cultures of mouse and human retinal and cardiac endothelial cells were exposed to protons, heavy ions, and gamma-rays. Alterations in protein expressions were dependent on radiation type, dose, and cell type. Doses =0.1 Gy caused a decrease in tubule network formation indicative of a reduced capacity for early-onset angiogenesis; 8) Mice were administered the radiation countermeasure gamma-tocotrienol for 4 weeks after oxygen ion irradiation. At 2 weeks, gamma-tocotrienol reduced cardiac protein levels of CD2 and collagen type III, but not of CD68 or mast cell tryptase. The analysis of cardiac function and cardiac tissue structure is ongoing.

Bibliography Type:	Description: (Last Updated: 01/08/2026)
Articles in Peer-reviewed Journals	Chang J, Luo Y, Wang Y, Pathak R, Sridharan V, Jones T, Mao XW, Nelson G, Boerma M, Hauer-Jensen M, Zhou D, Shao L. "Low doses of oxygen ion irradiation cause acute damage to hematopoietic cells in mice." PLoS One. 2016 Jul 1;11(7):e0158097. eCollection 2016. https://doi.org/10.1371/journal.pone.0158097 ; PubMed PMID: 27367604; PubMed Central PMCID: PMC4930193 , Jul-2016
Articles in Peer-reviewed Journals	Pathak R, Wang J, Garg S, Aykin-Burns N, Petersen KU, Hauer-Jensen M. "Recombinant thrombomodulin (Solulin) ameliorates early intestinal radiation toxicity in a preclinical rat model." Radiat Res. 2016 Aug;186(2):112-20. https://doi.org/10.1667/RR14408.1 ; PubMed PMID: 27459702; PubMed Central PMCID: PMC4995594 , Aug-2016
Articles in Peer-reviewed Journals	Wang Y, Boerma M, Zhou D. "Ionizing radiation-induced endothelial cell senescence and cardiovascular diseases." Radiat Res. 2016 Aug;186(2):153-61. Review. https://doi.org/10.1667/RR14445.1 ; PubMed PMID: 27387862; PubMed Central PMCID: PMC4997805 , Aug-2016
Articles in Peer-reviewed Journals	Boerma M. "An introduction to space radiation and its effects on the cardiovascular system." THREE. 2016 Oct 13:1-12. https://three.jsc.nasa.gov/articles/Boerma-Cardiovascular-and-IR.pdf , Oct-2016
Articles in Peer-reviewed Journals	Chang J, Wang Y, Pathak R, Sridharan V, Jones T, Mao XW, Nelson G, Boerma M, Hauer-Jensen M, Zhou D, Shao L. "Whole body proton irradiation causes acute damage to bone marrow hematopoietic progenitor and stem cells in mice." Int J Radiat Biol. 2017 Dec;93(12):1312-20. Epub 2017 Aug 7. https://doi.org/10.1080/09553002.2017.1356941 ; PubMed PMID: 28782442 , Dec-2017
Articles in Peer-reviewed Journals	Kiffer F, Carr H, Groves T, Anderson JE, Alexander T, Wang J, Seawright JW, Sridharan V, Carter G, Boerma M, Allen AR. "Effects of 1H + 16O charged particle irradiation on short-term memory and hippocampal physiology in a murine model." Radiat Res. 2018 Jan;189(1):53-63. https://doi.org/10.1667/RR14843.1 ; PubMed PMID: 29136391 , Jan-2018
Articles in Peer-reviewed Journals	Wang Y, Chang J, Li X, Pathak R, Sridharan V, Jones T, Mao XW, Nelson G, Boerma M, Hauer-Jensen M, Zhou D, Shao L. "Low doses of oxygen ion irradiation cause long-term damage to bone marrow hematopoietic progenitor and stem cells in mice." PLoS One. 2017 Dec 12;12(12):e0189466. eCollection 2017. https://doi.org/10.1371/journal.pone.0189466 ; PubMed PMID: 29232383; PubMed Central PMCID: PMC5726652 , Dec-2017
Articles in Peer-reviewed Journals	Carr H, Alexander TC, Groves T, Kiffer F, Wang J, Price E, Boerma M, Allen AR. "Early effects of 16O radiation on neuronal morphology and cognition in a murine model." Life Sciences in Space Research. 2018 May;17:63-73. Epub 2018 Mar 14. https://doi.org/10.1016/j.lssr.2018.03.001 ; PubMed PMID: 29753415 , May-2018
Articles in Peer-reviewed Journals	Mao XW, Boerma M, Rodriguez D, Campbell-Beachler M, Jones T, Stanbouly S, Sridharan V, Wroe A, Nelson GA. "Acute effect of low-dose space radiation on mouse retina and retinal endothelial cells." Radiat Res. 2018 Jul;190(1):45-52. Epub 2018 May 9. https://doi.org/10.1667/RR14977.1 ; PubMed PMID: 29741442 , Jul-2018

Articles in Peer-reviewed Journals	Kiffer F, Howe AK, Carr H, Wang J, Alexander T, Anderson JE, Groves T, Seawright JW, Sridharan V, Carter G, Boerma M, Allen AR. "Late effects of 1H irradiation on hippocampal physiology." Life Sci Space Res (Amst). 2018 May;17:51-62. Epub 2018 Mar 15. https://doi.org/10.1016/j.lssr.2018.03.004 ; PubMed PMID: 29753414 , May-2018
Articles in Peer-reviewed Journals	Koturbash I. "2017 Michael Fry Award Lecture. When DNA is actually not a target: Radiation epigenetics as a tool to understand and control cellular response to ionizing radiation." Radiat Res. 2018 Jul;190(1):5-11. Epub 2018 Apr 26. https://doi.org/10.1667/RR15027.1 ; PubMed PMID: 29697303; PubMed Central PMCID: PMC6036898 , Jul-2018
Articles in Peer-reviewed Journals	Miousse IR, Ewing LE, Kutanzi KR, Griffin RJ, Koturbash I. "DNA methylation in radiation-induced carcinogenesis: Experimental evidence and clinical perspectives." Crit Rev Oncog. 2018;23(1-2):1-11. Review. https://doi.org/10.1615/CritRevOncog.2018025687 ; PubMed PMID: 29953365; PubMed Central PMCID: PMC6369919 , Jul-2018
Articles in Peer-reviewed Journals	Cheema AK, Byrum SD, Sharma NK, Altadill T, Kumar VP, Biswas S, Balgley BM, Hauer-Jensen M, Tackett AJ, Ghosh SP. "Proteomic changes in mouse spleen after radiation-induced injury and its modulation by gamma-tocotrienol." Radiat Res. 2018 Nov;190(5):449-63. Epub 2018 Aug 2. https://doi.org/10.1667/RR15008.1 ; PubMed PMID: 30070965; PubMed Central PMCID: PMC6297072 , Nov-2018
Articles in Peer-reviewed Journals	Mao XW, Boerma M, Rodriguez D, Campbell-Beachler M, Jones T, Stanbouly S, Sridharan V, Nishiyama NC, Wroe A, Nelson GA. "Combined effects of low-dose proton radiation and simulated microgravity on the mouse retina and the hematopoietic system." Radiat Res. 2019 Sep;192(3):241-50. Epub 2018 Nov 15. https://doi.org/10.1667/RR15219.1 ; PubMed PMID: 30430917 , Sep-2019
Articles in Peer-reviewed Journals	Howe A, Kiffer F, Alexander TC, Sridharan V, Wang J, Ntagwabira F, Rodriguez A, Boerma M, Allen AR. "Long-term changes in cognition and physiology after low-dose 16O irradiation." Int J Mol Sci. 2019 Jan 7;20(1):E188. https://doi.org/10.3390/ijms20010188 ; PubMed PMID: 30621014; PubMed Central PMCID: PMC6337104 , Jan-2019
Articles in Peer-reviewed Journals	Seawright JW, Sridharan V, Landes RD, Cao M, Singh P, Koturbash I, Mao XW, Miousse IR, Singh SP, Nelson GA, Hauer-Jensen M, Boerma M. "Effects of low-dose oxygen ions and protons on cardiac function and structure in male C57BL/6J mice." Life Sciences in Space Research. 2019 Feb;20:72-84. https://doi.org/10.1016/j.lssr.2019.01.003 ; PubMed PMID: 30797436; PubMed Central PMCID: PMC6391741 , Feb-2019
Articles in Peer-reviewed Journals	Kiffer F, Alexander T, Anderson JE, Groves T, Wang J, Sridharan V, Boerma M, Allen AR. "Late effects of 16O-particle radiation on female social and cognitive behavior and hippocampal physiology." Radiat Res. 2019 Mar;191(3):278-94. Epub 2019 Jan 21. https://doi.org/10.1667/RR15092.1 ; PubMed PMID: 30664396 , Mar-2019
Articles in Peer-reviewed Journals	Kiffer F, Boerma M, Allen A. "Behavioral effects of space radiation: A comprehensive review of animal studies." Life Sci Space Res (Amst). 2019 May;21:1-21. Epub 2019 Feb 19. Review. https://doi.org/10.1016/j.lssr.2019.02.004 ; PubMed PMID: 31101151 , May-2019
Articles in Peer-reviewed Journals	Miousse IR, Skinner CM, Sridharan V, Seawright JW, Singh P, Landes RD, Cheema AK, Hauer-Jensen M, Boerma M, Koturbash I. "Changes in one-carbon metabolism and DNA methylation in the hearts of mice exposed to space environment-relevant doses of oxygen ions (16O)." Life Sci Space Res. 2019 Aug;22:8-15. https://doi.org/10.1016/j.lssr.2019.05.003 ; PMID: 31421852; PMCID: PMC6703167 , Aug-2019
Articles in Peer-reviewed Journals	Kiffer F, Alexander T, Anderson J, Groves T, McElroy T, Wang J, Sridharan V, Bauer M, Boerma M, Allen A. "Late effects of 1H + 16O on short-term and object memory, hippocampal dendritic morphology and mutagenesis." Front Behav Neurosci. 2020 Jun 26;14:96. https://doi.org/10.3389/fnbeh.2020.00096 ; PMID: 32670032; PMCID: PMC7332779 , Jun-2020
Articles in Peer-reviewed Journals	Sridharan V, Seawright JW, Landes RD, Cao M, Singh P, Davis CM, Mao XW, Singh SP, Zhang X, Nelson GA, Boerma M. "Effects of single-dose protons or oxygen ions on function and structure of the cardiovascular system in male Long Evans rats." Life Sci Space Res (Amst). 2020 Aug;26:62-8. https://doi.org/10.1016/j.lssr.2020.04.002 ; PMID: 32718688; PMCID: PMC7387753 , Aug-2020
Articles in Peer-reviewed Journals	Upadhyay M, Rajagopal M, Gill K, Li Y, Bansal S, Sridharan V, Tyburski JB, Boerma M, Cheema AK. "Identification of plasma lipidome changes associated with low dose space-type radiation exposure in a murine model." Metabolites. 2020 Jun 17;10(6):E252. https://doi.org/10.3390/metabo10060252 ; PMID: 32560360; PMCID: PMC7345467 , Jun-2020
Articles in Peer-reviewed Journals	Swinton C, Kiffer F, McElroy T, Wang J, Sridharan V, Boerma M, Allen AR. "Effects of 16O charged-particle irradiation on cognition, hippocampal morphology and mutagenesis in female mice." Behav Brain Res. 2021 Jun 11;407:113257. https://doi.org/10.1016/j.bbr.2021.113257 ; PMID: 33794227 , Jun-2021
Articles in Peer-reviewed Journals	Nemec-Bakk AS, Sridharan V, Landes RD, Singh P, Cao M, Seawright JW, Liu X, Zheng G, Dominic P, Pathak R, Boerma M. "Mitigation of late cardiovascular effects of oxygen ion radiation by gamma-tocotrienol in a mouse model." Life Sci Space Res (Amst). 2021 Nov;31:43-50. https://doi.org/10.1016/j.lssr.2021.07.006 ; PubMed PMID: 34689949; PubMed PMCID: PMC8548672 , Nov-2021
Articles in Peer-reviewed Journals	Nemec-Bakk AS, Sridharan V, Landes RD, Singh P, Cao M, Dominic P, Seawright JW, Chancellor JC, Boerma M. "Effects of low-dose oxygen ions on cardiac function and structure in female C57BL/6J mice." Life Sci Space Res (Amst). 2022 Feb;32:105-112. https://doi.org/10.1016/j.lssr.2021.12.004 ; PubMed PMID: 35065756; PubMed Central PMCID: PMC8803400 , Feb-2022
Articles in Peer-reviewed Journals	Nemec-Bakk AS, Sridharan V, Seawright JW, Nelson GA, Cao M, Singh P, Cheema AK, Singh B, Li Y, Koturbash I, Miousse IR, Ewing LE, Skinner CM, Landes RD, Lowery JD, Mao X-W, Singh SP, Boerma M. "Effects of proton and oxygen ion irradiation on cardiovascular function and structure in a rabbit model." Life Sci Space Res. 2023 May 1;37:78-87. https://doi.org/10.1016/j.lssr.2023.03.008 ; PMID: 37087182; PMCID: PMC10122719 , May-2023

Articles in Peer-reviewed Journals	Boerma M, Sridharan V, Mao XW, Nelson GA, Cheema AK, Koturbash I, Singh SP, Tackett AJ, Hauer-Jensen M. "Effects of ionizing radiation on the heart." Mutation Research. 2016 Oct - Dec;770(Pt B):319-27. Review. Epub 2016 Jul 10. https://doi.org/10.1016/j.mrrev.2016.07.003 ; PubMed PMID: 27919338; PubMed Central PMCID: PMC5144922 , Oct-2016
Articles in Peer-reviewed Journals	Casero D, Gill K, Sridharan V, Koturbash I, Nelson G, Hauer-Jensen M, Boerma M, Braun J, Cheema AK. "Space-type radiation induces multimodal responses in the mouse gut microbiome and metabolome." Microbiome. 2017 Aug 18;5(1):105. https://doi.org/10.1186/s40168-017-0325-z ; PubMed PMID: 28821301; PubMed Central PMCID: PMC5563039 , Aug-2017
Articles in Peer-reviewed Journals	Jayabalan GS, Wu YK, Bille JF, Kim S, Mao XW, Gimbel HV, Rauser ME, Fan JT. "In vivo two-photon imaging of retina in rabbits and rats." Exp Eye Res. 2018 Jan;166:40-48. Epub 2017 May 5. https://doi.org/10.1016/j.exer.2017.04.009 ; PubMed PMID: 28483661 , Jan-2018
Articles in Peer-reviewed Journals	Koturbash I, Merrifield M, Kovalchuk O. "Fractionated exposure to low doses of ionizing radiation results in accumulation of DNA damage in mouse spleen tissue and activation of apoptosis in a p53/Atm-independent manner." Int J Radiat Biol. 2017 Feb;93(2):148-55. Epub 2016 Oct 19. https://doi.org/10.1080/09553002.2017.1231943 ; PubMed PMID: 27758128 , Feb-2017
Articles in Peer-reviewed Journals	Mao XW, Nishiyama NC, Pecaut MJ, Campbell-Beachler M, Gifford P, Haynes KE, Becronis C, Gridley DS. "Simulated microgravity and low-dose/low-dose-rate radiation induces oxidative damage in the mouse brain." Radiation Research. 2016 Jun;185(6):647-57. https://doi.org/10.1667/RR14267.1 ; PubMed PMID: 27243749 , Jun-2016
Articles in Peer-reviewed Journals	Miousse IR, Chang J, Shao L, Pathak R, Nzabarushimana E, Kutanzi KR, Landes R, Tackett AJ, Hauer-Jensen M, Zhou D, Koturbash I. "Inter-strain differences in LINE-1 DNA methylation in the mouse hematopoietic system in response to exposure to low-doses of ionizing radiation." International Journal of Molecular Sciences. 2017 Jul 4;18(7):E1430. https://doi.org/10.3390/ijms18071430 ; PubMed PMID: 28677663; PubMed Central PMCID: PMC5535921 , Jul-2017
Articles in Peer-reviewed Journals	Miousse IR, Kutanzi KR, Koturbash I. "Effects of ionizing radiation on DNA methylation: from experimental biology to clinical applications." International Journal of Radiation Biology. 2017 May;93(5):457-69. Review. Epub 2017 Feb 21. https://doi.org/10.1080/09553002.2017.1287454 ; PubMed PMID: 28134023; PubMed Central PMCID: PMC5411327 , May-2017
Articles in Peer-reviewed Journals	Miousse IR, Tobacyk J, Melnyk S, James SJ, Cheema AK, Boerma M, Hauer-Jensen M, Koturbash I. "One-carbon metabolism and ionizing radiation: a multifaceted interaction." Biomolecular Concepts. 2017 May 24;8(2):83-92. Review. https://doi.org/10.1515/bmc-2017-0003 ; PubMed PMID: 28574375 , May-2017
Articles in Peer-reviewed Journals	Pathak R, Bachri A, Ghosh SP, Koturbash I, Boerma M, Binz RK, Sawyer JR, Hauer-Jensen M. "The vitamin E analog gamma-tocotrienol (GT3) suppresses radiation-induced cytogenetic damage." Pharmaceutical Research. 2016 Sep;33(9):2117-25. Epub 2016 May 23. https://doi.org/10.1007/s11095-016-1950-0 ; PubMed PMID: 27216753; PubMed Central PMCID: PMC4967083 , Sep-2016
Articles in Peer-reviewed Journals	Pathak R, Koturbash I, Hauer-Jensen M. "Detection of inter-chromosomal stable aberrations by multiple fluorescence in situ hybridization (mFISH) and spectral karyotyping (SKY) in irradiated mice." Journal of Visualized Experiments. 2017 Jan 11;(119):e55162. https://doi.org/10.3791/55162 ; PubMed PMID: 28117817; PubMed Central PMCID: PMC5352253 , Jan-2017
Articles in Peer-reviewed Journals	Prior S, Miousse IR, Nzabarushimana E, Pathak R, Skinner C, Kutanzi KR, Allen AR, Raber J, Tackett AJ, Hauer-Jensen M, Nelson GA, Koturbash I. "Densely ionizing radiation affects DNA methylation of selective LINE-1 elements." Environmental Research. 2016 Oct;150:470-81. Epub 2016 Jul 14. https://doi.org/10.1016/j.envres.2016.06.043 ; PubMed PMID: 27419368; PubMed Central PMCID: PMC5003736 , Oct-2016
Articles in Peer-reviewed Journals	Seawright JW, Samman Y, Sridharan V, Mao XW, Cao M, Singh P, Melnyk S, Koturbash I, Nelson GA, Hauer-Jensen M, Boerma M. "Effects of low-dose rate gamma-irradiation combined with simulated microgravity on markers of oxidative stress, DNA methylation potential, and remodeling in the mouse heart." PLoS One. 2017 Jul 5;12(7):e0180594. eCollection 2017. https://doi.org/10.1371/journal.pone.0180594 ; PubMed PMID: 28678877; PubMed Central PMCID: PMC5498037 , Jul-2017
Awards	Tackett A. (Alan Tackett) "Scharlau Family Endowed Chair, December 2016." Dec-2016
Awards	Boerma M. (Marjan Boerma) "Radiation Research Society J.W. Osborne Award, October 2016." Oct-2016
Awards	Hauer-Jensen M. (Martin Hauer-Jensen) "J. Thomas May Distinguished Endowed Chair, August 2016." Aug-2016